

Flood Susceptibility Mapping for Elmanagil, Sudan

A Screening-Level Weighted Overlay Using Terrain, Drainage, Rainfall, and Land Cover Indicators

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Objective

The project presents a screening-level flood susceptibility map for Elmanagil locality, Sudan, part of the Gezira agricultural scheme in the Eastern Nile sub-basin. The analysis combines terrain, drainage, seasonal rainfall, and land cover indicators into a single weighted overlay, in the absence of a detailed local irrigation-canal network dataset. It was carried out as a follow-on to a prior rainfall products comparison study over the same area.

Study area:

Elmanagil locality lies within the Gezira agricultural scheme, Sudan, an area where rainfall availability directly affects supplementary rainfed agriculture and irrigation planning. the study area boundary was uploaded as an asset in Google Earth Engine where the rest of the analysis was performed.

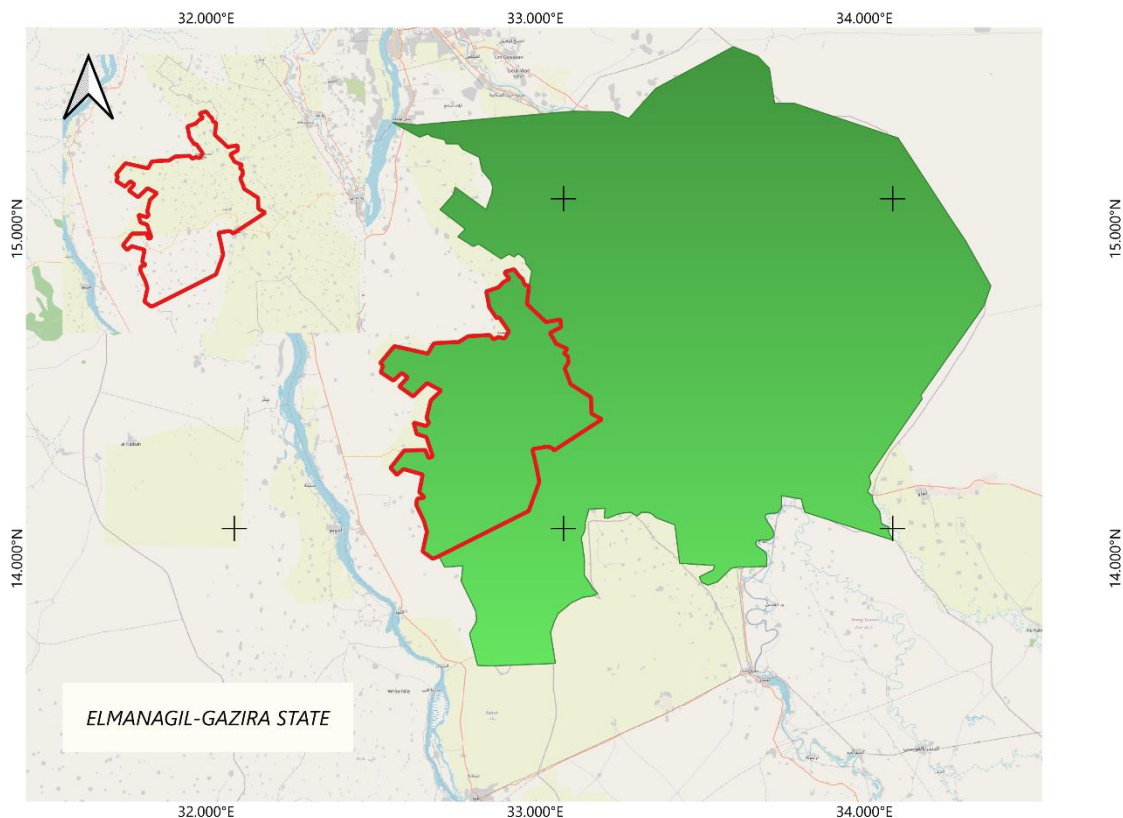


Figure 1. Study area boundary (Elmanagil)

Data and Methodology

Indicators used

- HAND – Height Above Nearest Drainage (MERIT Hydro, ~90 m): vertical proximity to the nearest natural drainage line
- Flow accumulation (MERIT Hydro): upstream contributing area, indicating natural water-concentration pathways
- Slope (SRTM DEM): terrain steepness, with flatter terrain favoring water accumulation
- Seasonal (JJAS) rainfall (CHIRPS, 2001–2023 mean): reused from the companion rainfall comparison study
- Land cover (ESA WorldCover): surface type, reclassified into a risk scale (e.g., water and wetland classes scored highest, bare soil and vegetated natural cover scored lowest)

Processing steps

- Extracted HAND, flow accumulation, and slope for the study area in Google Earth Engine
- Reused the JJAS seasonal rainfall layer computed for the companion rainfall comparison study
- Reclassified land cover into a 1–5 risk scale based on surface type
- Normalized each continuous layer (HAND, flow accumulation, slope, rainfall) to a 1–5 risk score using min-max scaling within the study area boundary, inverting HAND and slope so that lower raw values (closer to drainage, flatter terrain) map to higher risk scores
- Combined all five layers into a single susceptibility map using a weighted linear overlay

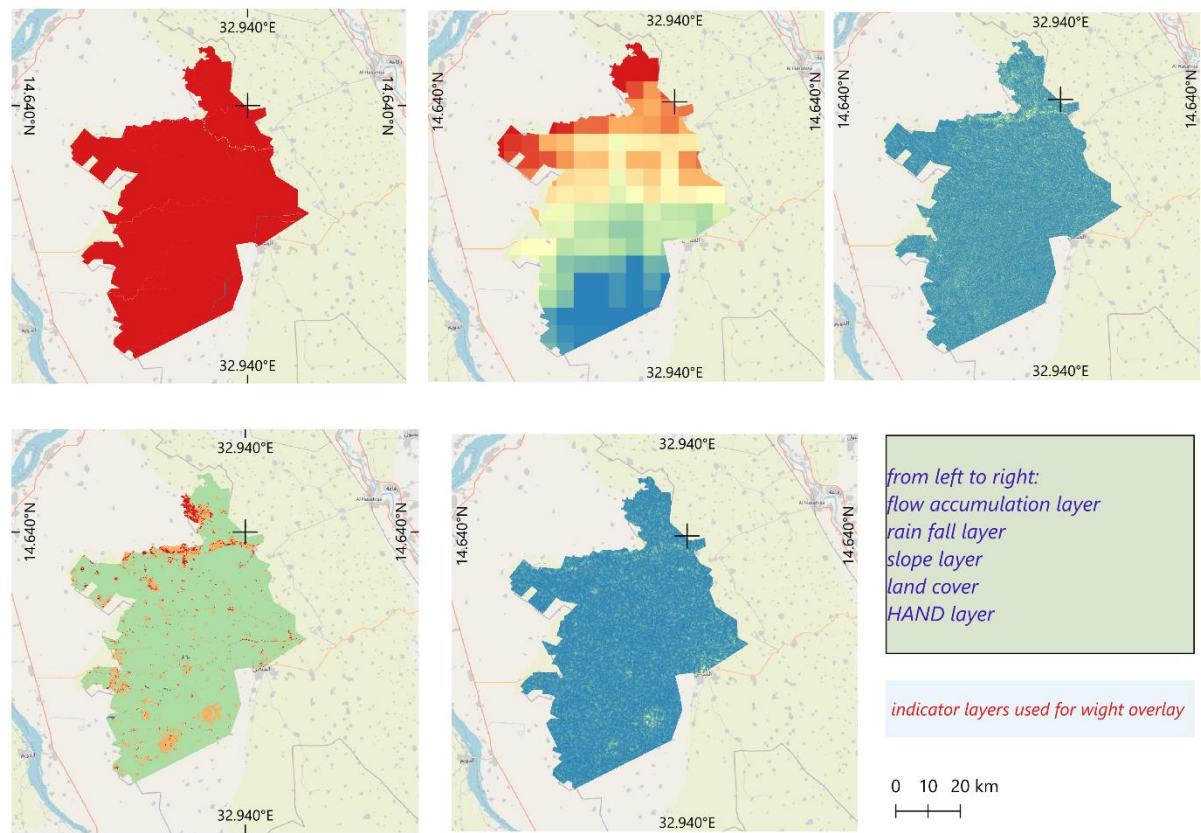
Overlay weights

The following weights were assigned based on the general flood-risk literature rather than calibration against observed flood events, and should be treated as a first-pass assumption:

Indicator	Weight	Rationale
HAND	35%	Most direct indicator of vertical proximity to drainage
Flow accumulation	20%	Captures natural water-concentration pathways
Slope	20%	Flat terrain favors ponding and slow drainage
Seasonal rainfall	15%	Wetter areas face greater flood loading

Indicator	Weight	Rationale
Land cover	10%	Surface type affects infiltration and runoff

Figure 2. Individual risk layers – HAND, flow accumulation, slope, rainfall, land cover



Results:

Individual risk layers

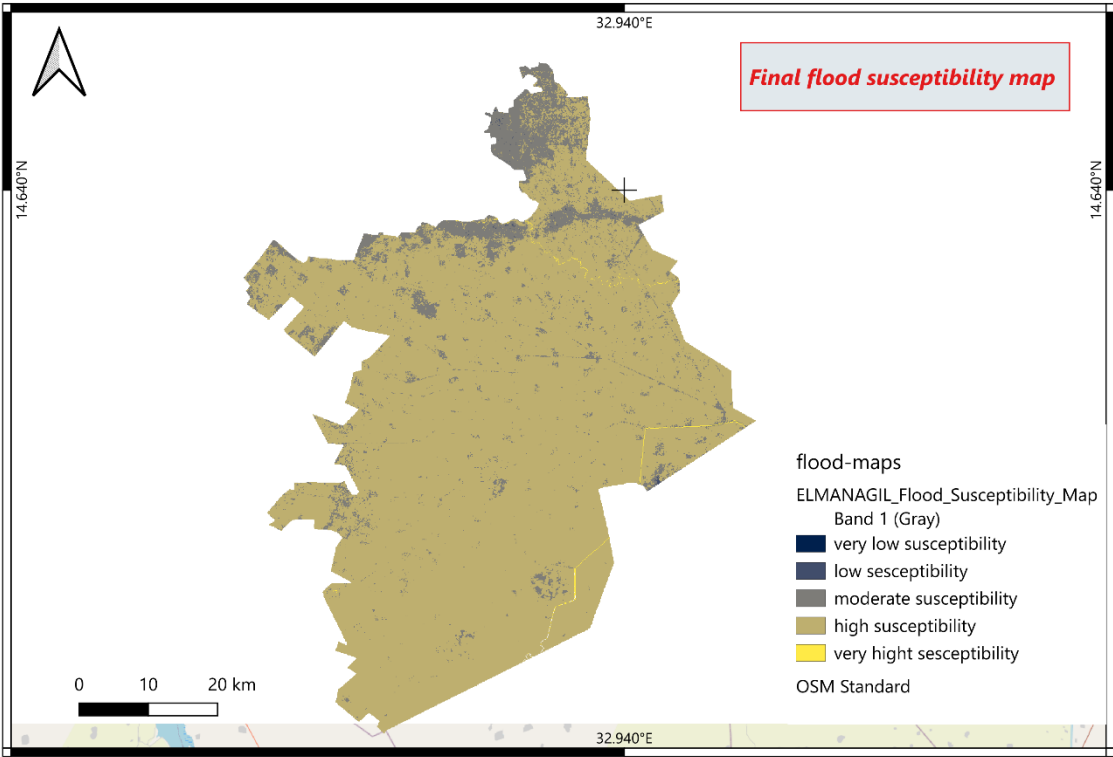
Each indicator was mapped independently as a 1–5 risk score before combination, allowing visual inspection of which factor drives risk in different parts of the study area.

Final flood susceptibility map

The combined susceptibility score ranged from 2.34 to 4.90 (scale of 1–5), with a study-area mean of 3.76. This indicates that most of Elmanagil falls into the medium-to-high susceptibility range, consistent with its setting as a flat floodplain near the Blue Nile, rather than showing a sharp contrast between clearly “safe” and “at-risk” zones.

Statistic	Value (1–5 scale)
Minimum	2.34
Maximum	4.90
Mean	3.76

Figure 3. Final flood susceptibility map (weighted overlay)



Discussion

The relatively narrow overall range (2.34–4.90) combined with a high mean (3.76) suggests that flood susceptibility across Elmanagil is generally elevated and fairly uniform, rather than concentrated in a few isolated hotspots. This is consistent with the area's physical setting: a low-relief, intensively irrigated plain close to the Blue Nile, where small differences in terrain and drainage proximity produce moderate rather than extreme variation in susceptibility. Areas with comparatively lower scores tended to correspond to slightly elevated or better-drained terrain, while higher scores aligned with lower-lying areas and locations with greater flow accumulation.

Limitations

- A local irrigation-canal network dataset was not available; HAND and flow accumulation from MERIT Hydro (~90 m resolution) were used as a substitute. This captures natural drainage patterns but may not fully represent engineered irrigation and drainage infrastructure specific to the Gezira scheme.
- Overlay weights are assumption-based and have not been calibrated against observed flood events, local knowledge, or expert elicitation.
- The 90 m resolution of MERIT Hydro may not resolve small-scale drainage features relevant at the field or canal scale.
- This is an exploratory, screening-level analysis intended to identify relatively more and less susceptible areas; it is not a substitute for full hydraulic or hydrodynamic modelling (e.g., HEC-RAS) for operational flood-risk mapping or engineering design.

Conclusion:

The analysis indicates that flood susceptibility across Elmanagil is generally moderate to high, driven primarily by low relief and proximity to natural drainage pathways rather than by sharply localized terrain features. Where feasible, future work should prioritize obtaining a local canal and drainage network dataset and any available historical flood-extent or damage records to calibrate the overlay weights and replace the MERIT Hydro-based approximation with locally validated drainage information. This would move the analysis from a screening-level tool toward one suitable for operational flood-risk planning in the Gezira scheme.

References

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